Rebuttal Figures for "L-CUBE: Isolating Long-Context Capacity from Knowledge with Controllable Mutual Information Scaling"

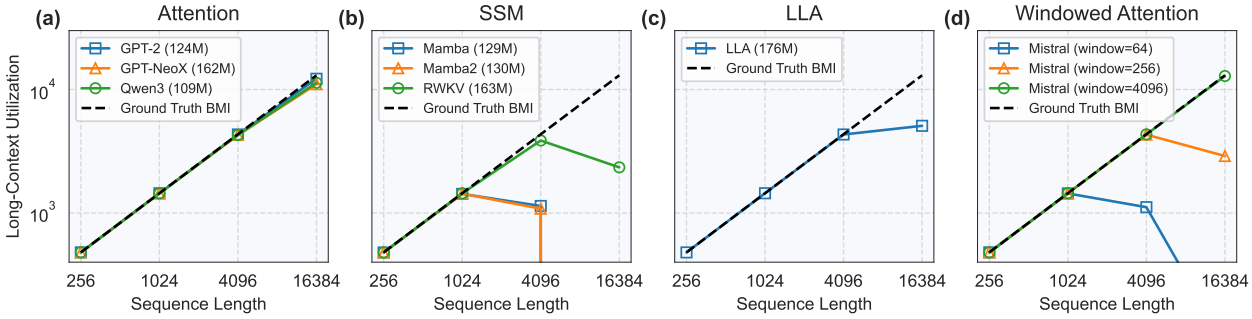

Figure R1: Utilization curve at iteration 150k. While the utilization for all models are lower compared to the results at the end of training, the general behaviors and trends are the same.

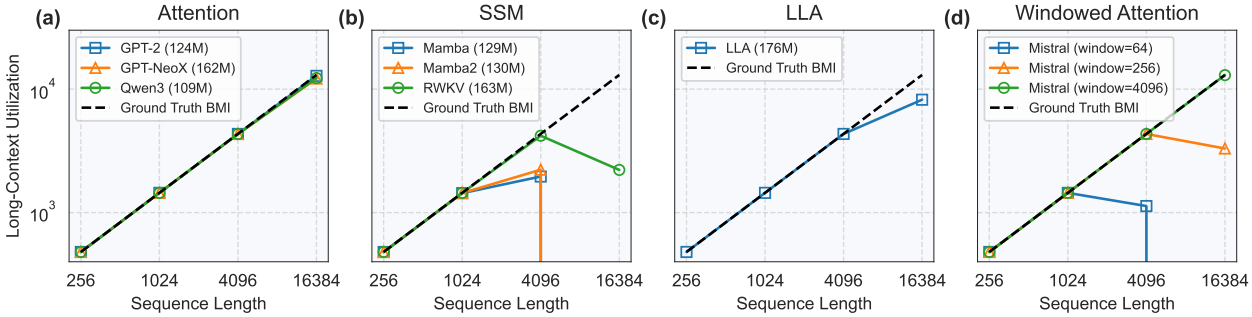

Figure R2: Utilization curve at iteration 250k.While the utilization for all models are lower compared to the results at the end of training, the general behaviors and trends are the same.

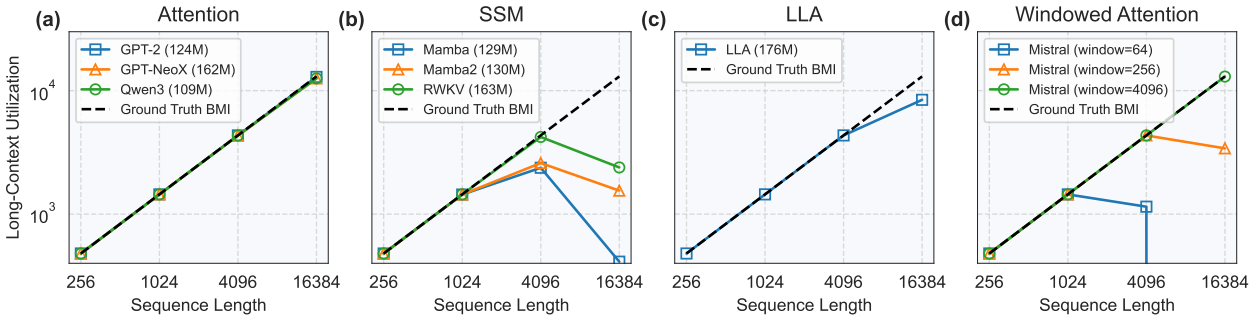

Figure R3: Utilization curve at iteration 350k. The uiltizations for all models are very close to the results at the end of training, except Mamba and Mamba2.

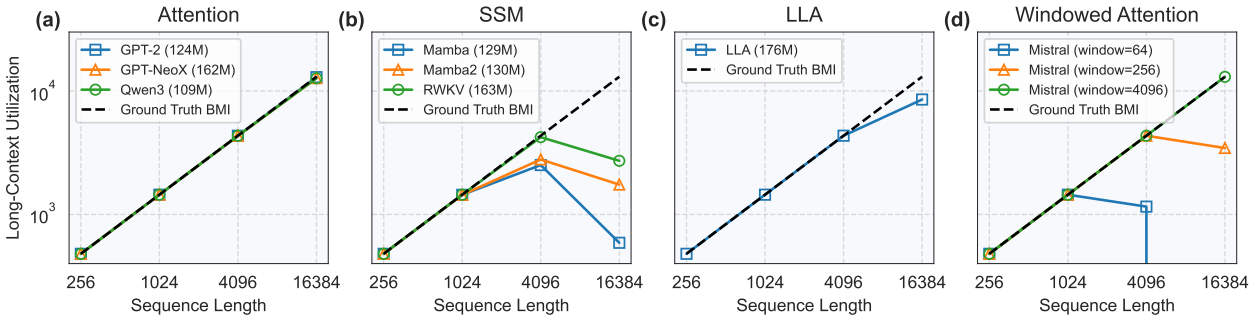

Figure R4: Utilization curve at iteration 500k (end of training). This is the same figure shown in the main text of the paper.

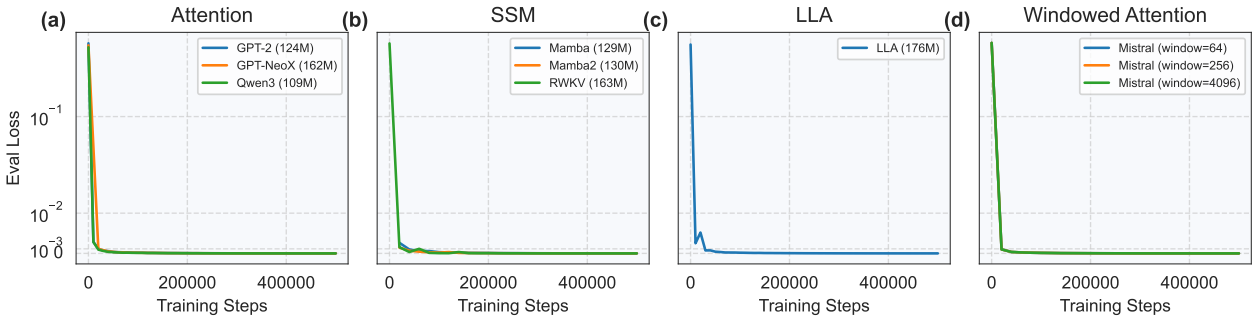

Figure R5: Training curves for sequence length = 64. the training converges well before the end of training. All models perform well at this sequence length.

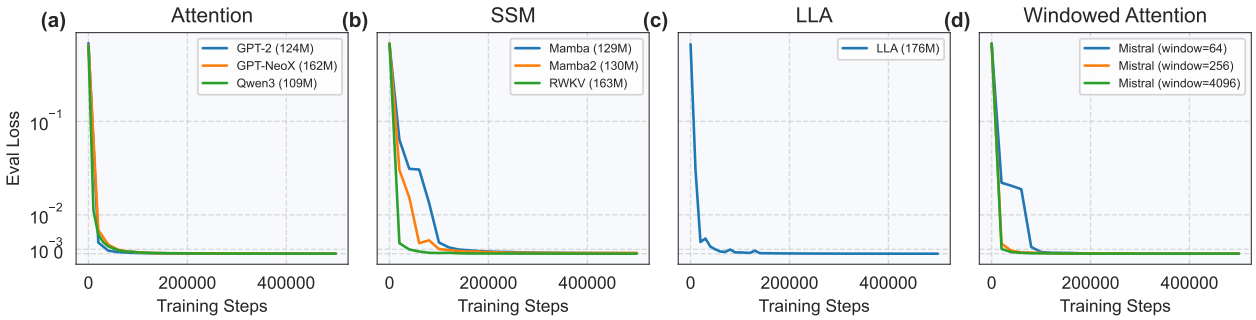

Figure R6: Training curves for sequence length = 1024. the training converges well before the end of training. All models still perform similarlywell at this sequence length.

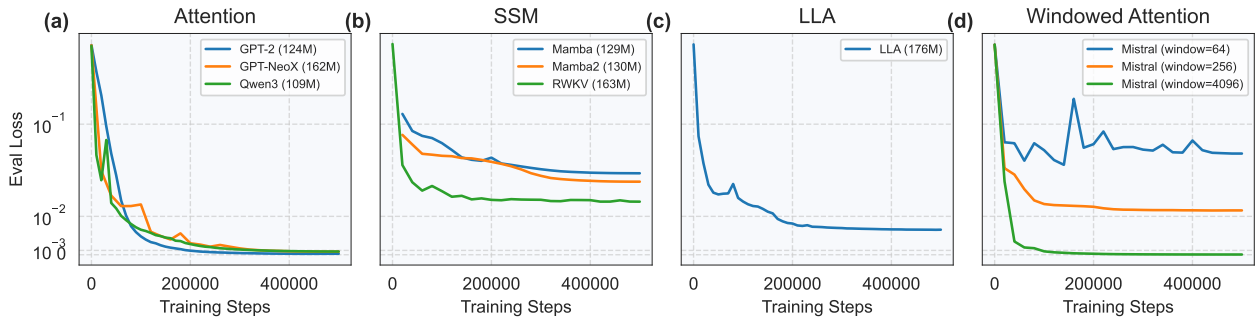

Figure R7: Training curves for sequence length = 16384. The seperation between models becomes clear at this sequence length, but it is clear that all models converges well before the end of training.

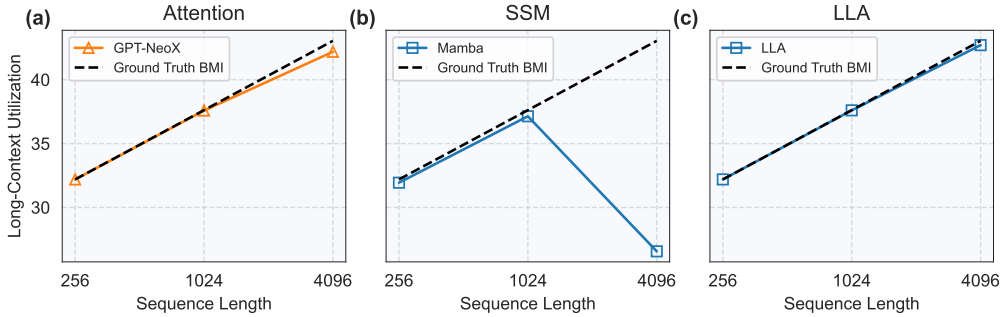

Figure R8: Utilization curves for GPT-NeoX, Mamba, and log-linear attention as representitive models for vanilla attention, state space model, and log-linear-attention. Here, the generator is tuned to have a logarithmic scaling of the bipartite mutual information. It is clear from the figure that in this setting, both vanilla attention and the log-linear attention mechanism are capable of represnting the logarithmically increasing bipartite mutual information. where as the state space model, only capable of representing a constant bipartite mutual information fails at long sequence lengths.

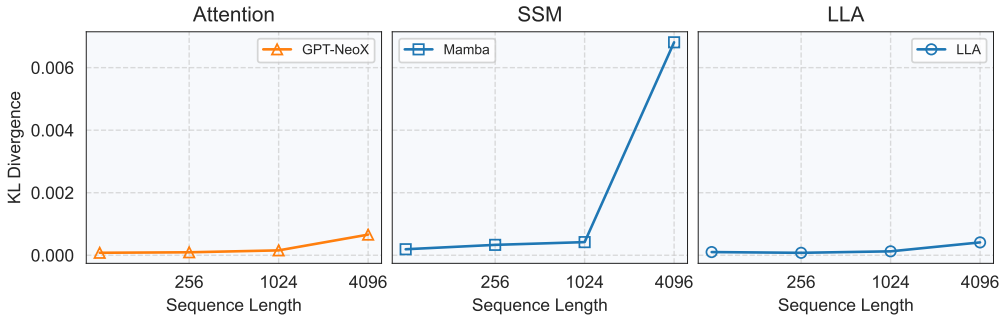

Figure R9: Raw KL divergence averaged over tokens for GPT-NeoX, Mamba, and log-linear-attention as representitive models for vanilla attention, state space model, and log-linear-attention. This figure shows similar trends as Fig. R8.

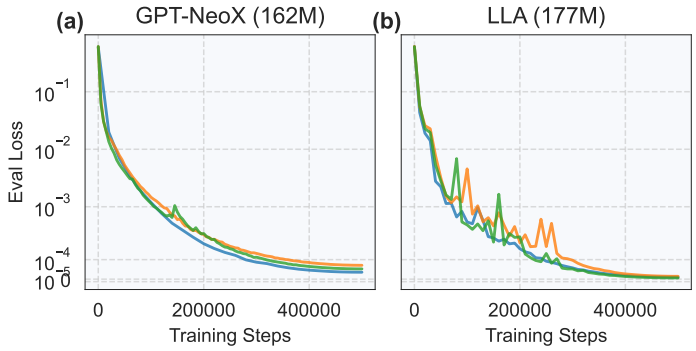

Figure R10: Training curves for different seeds at sequence length = 4096 for representative models. This figure shows the stability of the training across different seeds.